

FROM MRI TO BIOPSY: CONTRAST-ENHANCED DIGITAL MAMMOGRAPHY AS A DIAGNOSTIC BRIDGE FOR OCCULT CONTRALATERAL BREAST LESIONS

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BACKGROUND

Breast MRI frequently detects additional lesions that are occult on mammography and ultrasound. When MRI-guided biopsy is unavailable, contrast-enhanced digital mammography (CEDM) may provide an alternative for lesion localisation and image-guided biopsy. This case highlights the role of CEDM in diagnosing an occult synchronous contralateral breast cancer and its impact on patient management

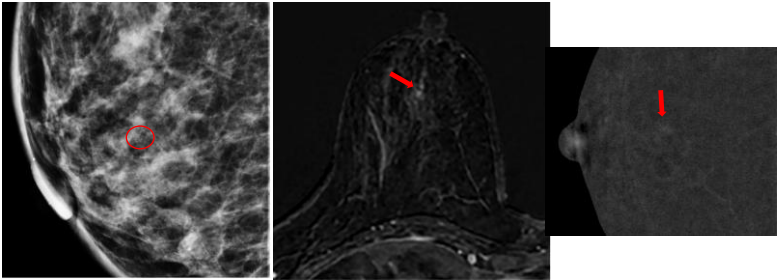
OBJECTIVES

- To demonstrate the role of CEDM in localising and guiding biopsy of MRI-detected non-mass enhancement (NME) without mammographic or sonographic correlation.
- To illustrate imaging–pathology correlation of an occult contralateral breast lesion detected on MRI and CEDM.
- To highlight a practical diagnostic pathway when MRI-guided breast intervention is unavailable.

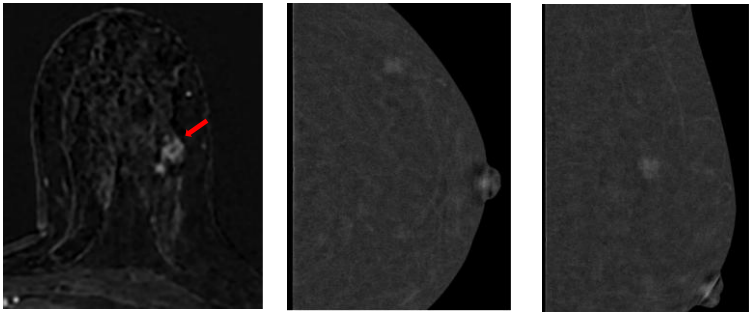
CASE PRESENTATION

A 59-year-old woman presented with a self-detected right breast lump. Mammography showed suspicious grouped right breast calcifications, and stereotactic biopsy confirmed carcinoma in situ (DCIS). Preoperative MRI detected an additional focal non-mass enhancement (NME) in the left upper outer quadrant without a mammographic or sonographic correlate. CEDM demonstrated corresponding moderate enhancement. CEDM-guided biopsy with twirl marker placement confirmed synchronous invasive carcinoma.

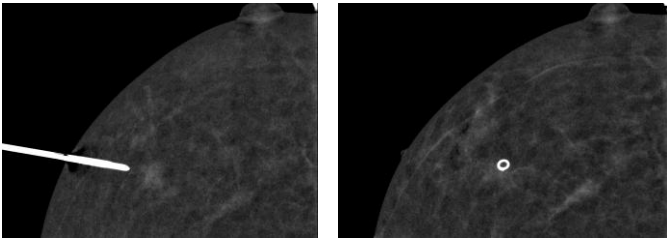
IMAGING FINDINGS



MMG: Grouped amorphous calcifications, right retroareolar MRI: Linear clumped NME, right retroareolar region with corresponding focus mild enhancement on CEDM



Left UOQ focal NME on MRI CEDM shows corresponding moderate enhancement in the left UOQ



CEDM-guided biopsy with twirl marker placement

CONCLUSION / TAKE-HOME MESSAGE

CEDM provides a practical problem-solving tool for MRI-detected breast lesions without mammographic or sonographic correlation. When the lesion can be reproduced on CEDM, CEDM-guided biopsy enables tissue diagnosis and may avoid the need for MRI-guided intervention.

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DIAGNOSTIC PATHWAY

Right DCIS prompted MRI, which detected occult left UOQ NME. CEDM confirmed enhancement and guided biopsy revealing synchronous invasive carcinoma.

Right DCIS

MRI: occult left NME

CEDM localisation

Biopsy: invasive carcinoma